MIL-A-22434A(Wep)
7 March 1961

SUPERSEDING
MIL-A-22434(Wep)
10 February 1960

MILITARY SPECIFICATION

ADHESIVE, POLYESTER, THIXOTROPIC

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements for the procurement of a thixotropic polyester adhesive for use in bonding glass cloth epoxy resin laminate rings to metal components of warhead assemblies where special application and assembly techniques are involved.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications of the issue in effect on the date of invitation for bids form a part of this specification.

SPECIFICATIONS

FEDERAL

TT-D-643	Drier, Paint, Naphthenate, Liquid Concentrated

MILITARY

MIL-P-3863	Packaging of Chemicals, Liquid
MIL-G-21729(Wep)	Glass-Fiber Base Laminate, Epoxy Resin

FSC 8010

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

OS 8819	Resin, Alkyd-Styrene, Thermosetting

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM Designation: E 4-55T	Verification of Testing Machines (Tentative)

(Copies may be purchased directly from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

3. REQUIREMENTS

3.1 Description.- The thixotropic adhesive is a homogeneous compound prepared by combining colloidal silica with a liquid unsaturated alkyd-styrene type thermosetting resin. Methyl-ethyl ketone peroxide is used as the catalyst and cobalt naphthenate as the accelerator to polymerize the resin.

3.2 Materials.- The materials used in formulating this adhesive shall conform to the specifications listed in 2.1. In addition, the following shall apply (See 6.3 for advisory information).

3.2.1 Liquid resin.- The liquid resin used shall conform to the requirements of OS 8819, Class 1 and Class 2.

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3.2.2 Thixotropic agent.- The thixotropic agent shall be a finely divided silicon dioxide of a minimum of 99% purity having a free moisture content of 2% maximum and a particle size range of 0.015 to 0.020 microns.

3.2.3 Catalyst.- The catalyst shall be a clear, colorless solution of methyl-ethyl ketone peroxide in dimethyl-phthalate. This mixture shall have an active oxygen content of 11.0 percent minimum, specific gravity of 1.093 minimum, flash point (Cleveland Open Cup) above 50°C, and a refractive index of 1.4530 ± .0010.

3.2.4 Accelerator.- The accelerator shall be cobalt naphthenate conforming to the requirements of Federal Specification TT-D-643, Type II.

3.3 Composition.- The composition of the adhesive shall be in accordance with Table I.

TABLE I

Ingredients	Parts by weight
Resin,Class 1	75.0 ± 3.75
Resin,Class 2	25.0 ± 1.25
Accelerator	0.8 ± 0.04
Catalyst	1.5 ± 0.075
Thixotropic agent	6.0 ± 1.0

3.4 Mechanical and physical properties.- The mechanical and physical properties of the adhesive shall be determined in accordance with tests detailed herein.

3.4.1 Tensile strength.- The minimum average tensile strength of the adhesive shall be 3000 pounds per square inch when cured in accordance with 4.4.3.4 (a) or 2500 pounds per square inch when cured in accordance with 4.4.3.4.(b).

3.4.2 Viscosity.- The viscosity of the unpolymerized resin formulated as described in 4.4.2 shall be determined in accordance with 4.4.4. As illustrated in Figure 4, acceptable material shall maintain an irregular surface, forming peaks after extrusion thru an orifice.

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4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Production batch.- For purpose of sampling and inspection, a production batch shall be limited to the quantity of adhesive which may be prepared from the same lots or batches of raw materials used in preparation of the preproduction samples.

4.3 Sampling and inspection.

4.3.1 Production batch acceptance test procedure.- Acceptance of production batch shall be based on results of test data derived from preproduction test specimens. Preproduction test specimens are prepared by the fabricator in advance of the production batch using the same formulation and processing procedure as for the production batch. Test results from preproduction test specimens shall meet the requirements of this specification and shall be approved by the procuring agency or the Government inspector prior to the manufacture of each production batch.

4.4 Test procedure.

4.4.1 Conditioning room or desiccators.- A conditioning room capable of maintaining a relative humidity of 50 ± 2% at 73.4° F ± 2° F or desiccators filled with a saturated salt solution to give a relative humidity of 50 ± 2% at 73.4° F ± 2° F shall be used for conditioning test specimens. A saturated salt solution of calcium nitrate will give approximately 51% relative humidity at 75° F.

4.4.2 Formulation.

4.4.2.1 Material.- The resin, thixotropic agent, catalyst and accelerator shall conform to the requirements of 3.2.

4.4.2.2 Mixing procedure.- Mix 75.0 ± 3.75 parts by weight of Class 1 resin and 25.0 ± 1.25 parts by weight of Class 2 resin. Add 0.8 ± 0.04 parts of the accelerator and mix thoroughly. After the accelerator has been

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thoroughly mixed with the resin add 1.5 ± 0.075 parts by weight of the catalyst. WARNING: In no case should the accelerator, cobalt naphthenate, be mixed directly with the catalyst, methyl-ethyl ketone peroxide. Such a mixture may decompose violently. After the above ingredients have been thoroughly mixed, add 6.0 ± 1.0 parts by weight of the thixotropic agent which has previously been prepared by drying at 110°C ± 1.0°C for 24 hours and desiccated until used.

4.4.3 Tensile strength.- The tensile strength of the adhesive shall be determined as the maximum load per unit area, applied normal to the glue line, carried by a joint of the type described herein and tested according to the procedure indicated herein. A minimum of five specimens shall be used.

4.4.3.1 Apparatus.- The apparatus shall consist of the following:

4.4.3.1.1 Testing machine.- A testing machine capable of maintaining a specified rate of loading and in which the error for indicated loads which are to be measured shall not exceed plus or minus one percent. The load indicating mechanism shall be essentially free from inertia lag at a specified rate of testing. The accuracy of the testing machine shall be verified in accordance with the tentative method of verification of testing machines (A.S.T.M. Designation E-4-55T). The testing machine shall consist essentially of a fixed member carrying one grip and a movable member carrying a second grip. The grips for holding a test specimen between the fixed member and the movable member shall be of the self-aligning type. That is, they shall be attached to the fixed and movable members in such a way that they will move into alignment as soon as any load is applied so that the long axis of the test specimen will coincide with the direction of the applied load. A design for grips which has proved satisfactory is shown in Figure 1.

4.4.3.2 Test specimen.- The test specimen shall conform to the form and dimensions indicated in Figure 2.

4.4.3.2.1 Machining of Part A.- The two adjacent sides shall be ground so that the included angle is 90° ± 0°2'. The ends are to be ground perpendicular to the ground sides so that the included angle is 90° ± 0°2'. The drilled hole is to be perpendicular to one ground side so that the included angle is 90° ± 0°2'. The drilled hole must always be in the same relative position to the ground surfaces as indicated in Figure 2. The initial finish on surfaces to be bonded shall be 16 micro inches or better.

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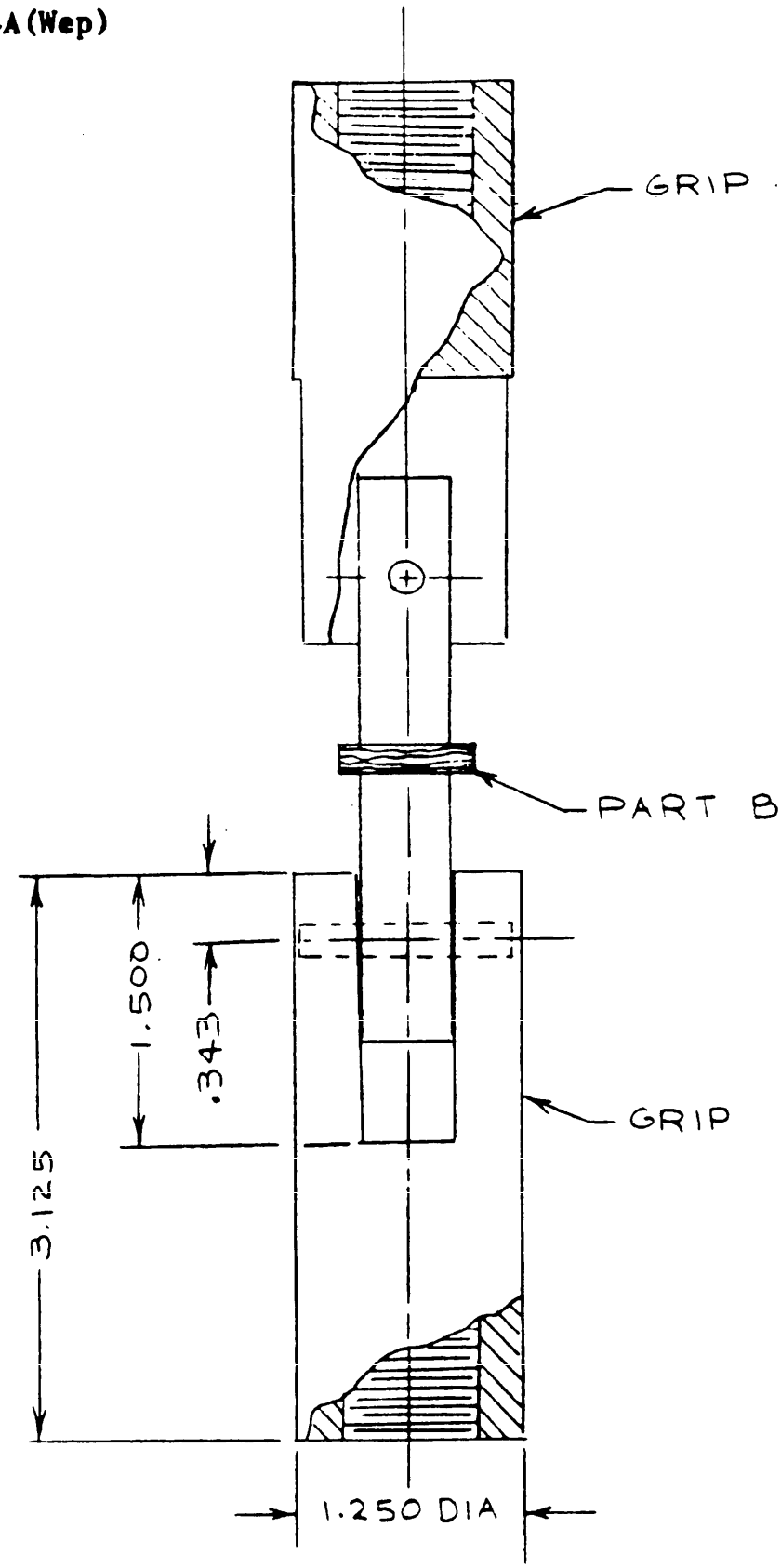
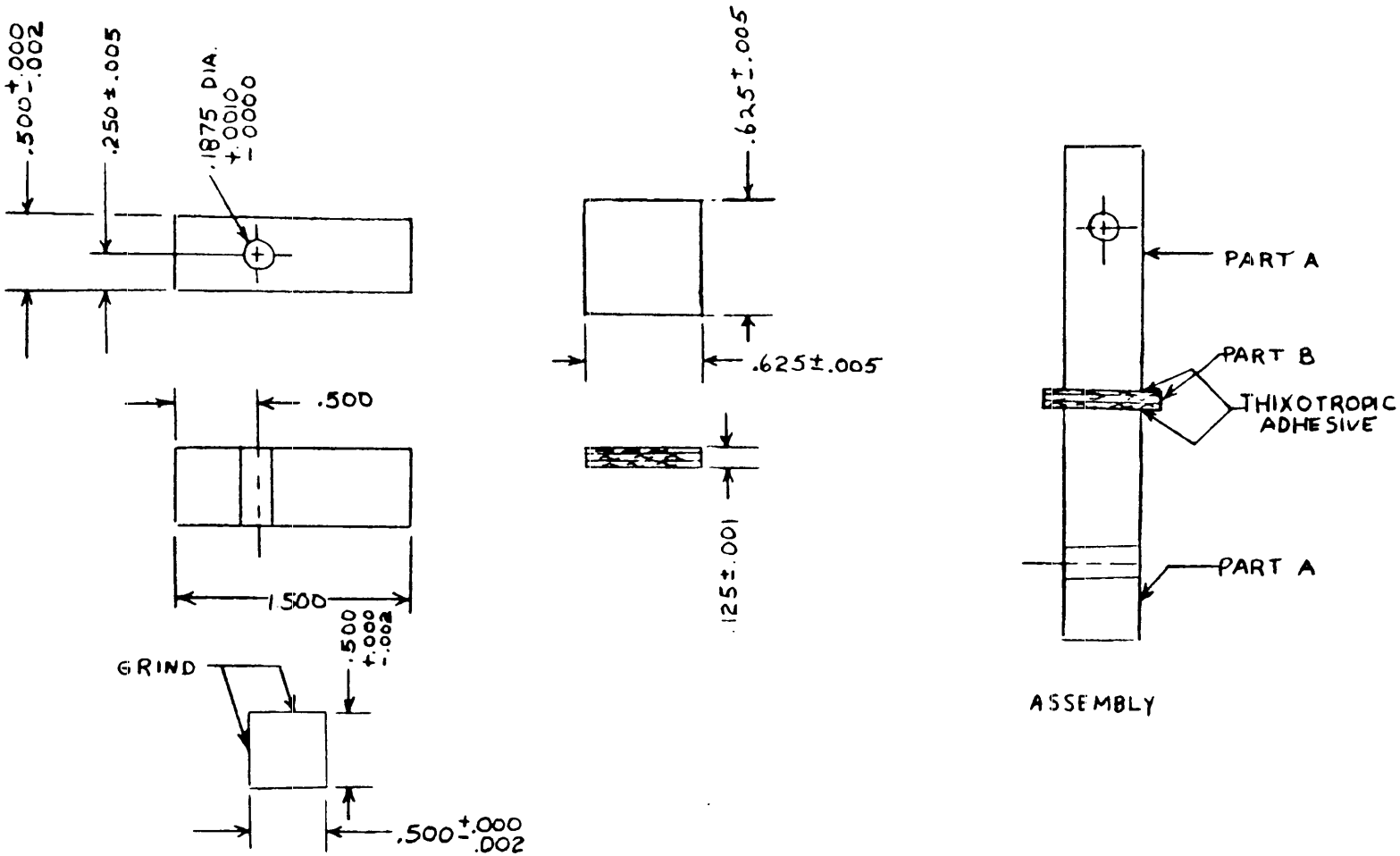


FIG. 1 TENSILE SPECIMEN AND GRIPS

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PART A- STAINLESS STEEL 18-8, TYPE 303

PART B- GLASS CLOTH-EPOXY RESIN LAMINATE

TEST SPECIMENS FOR TENSILE STRENGTH OF ADHESIVE

Figure 2

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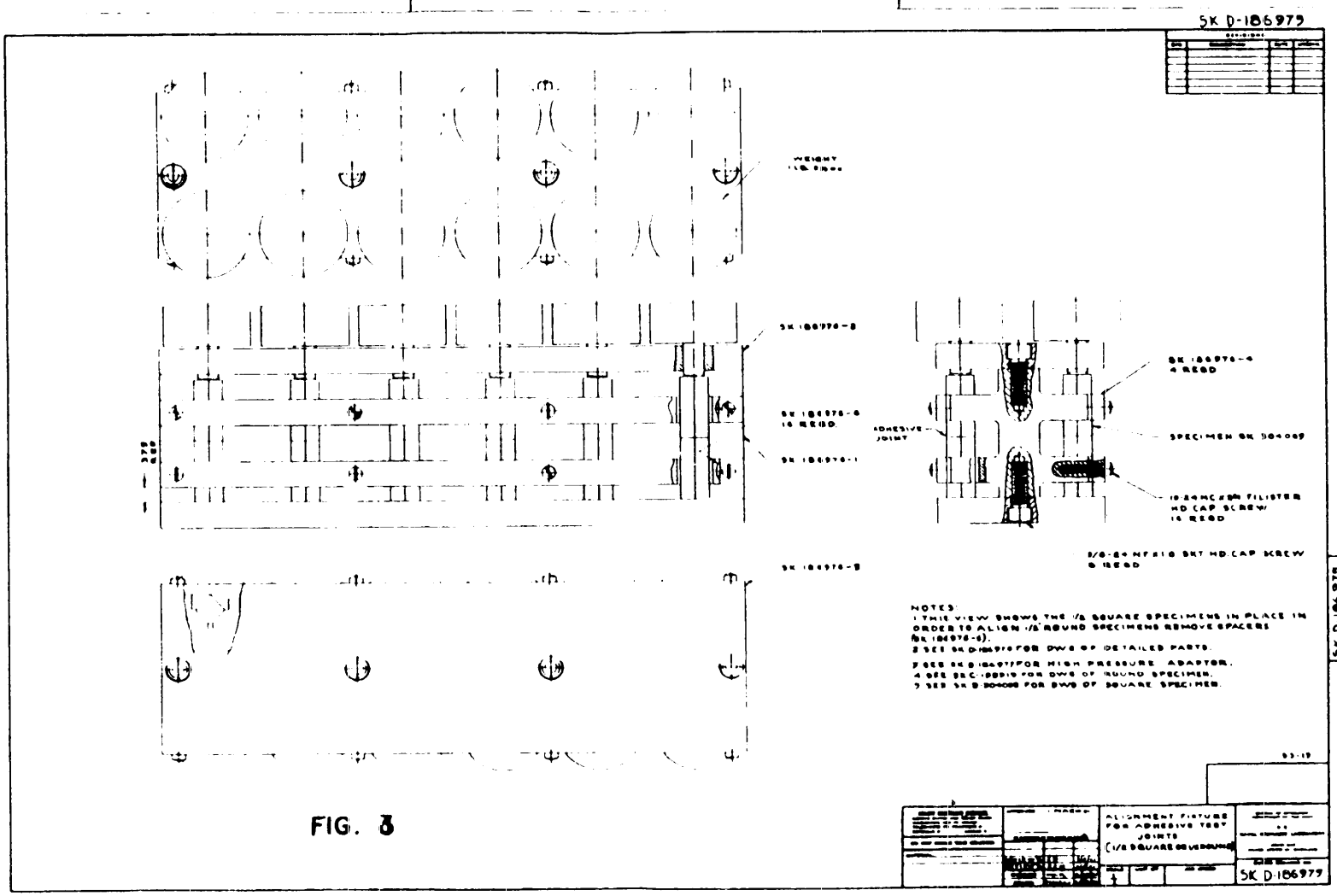


FIG. 3

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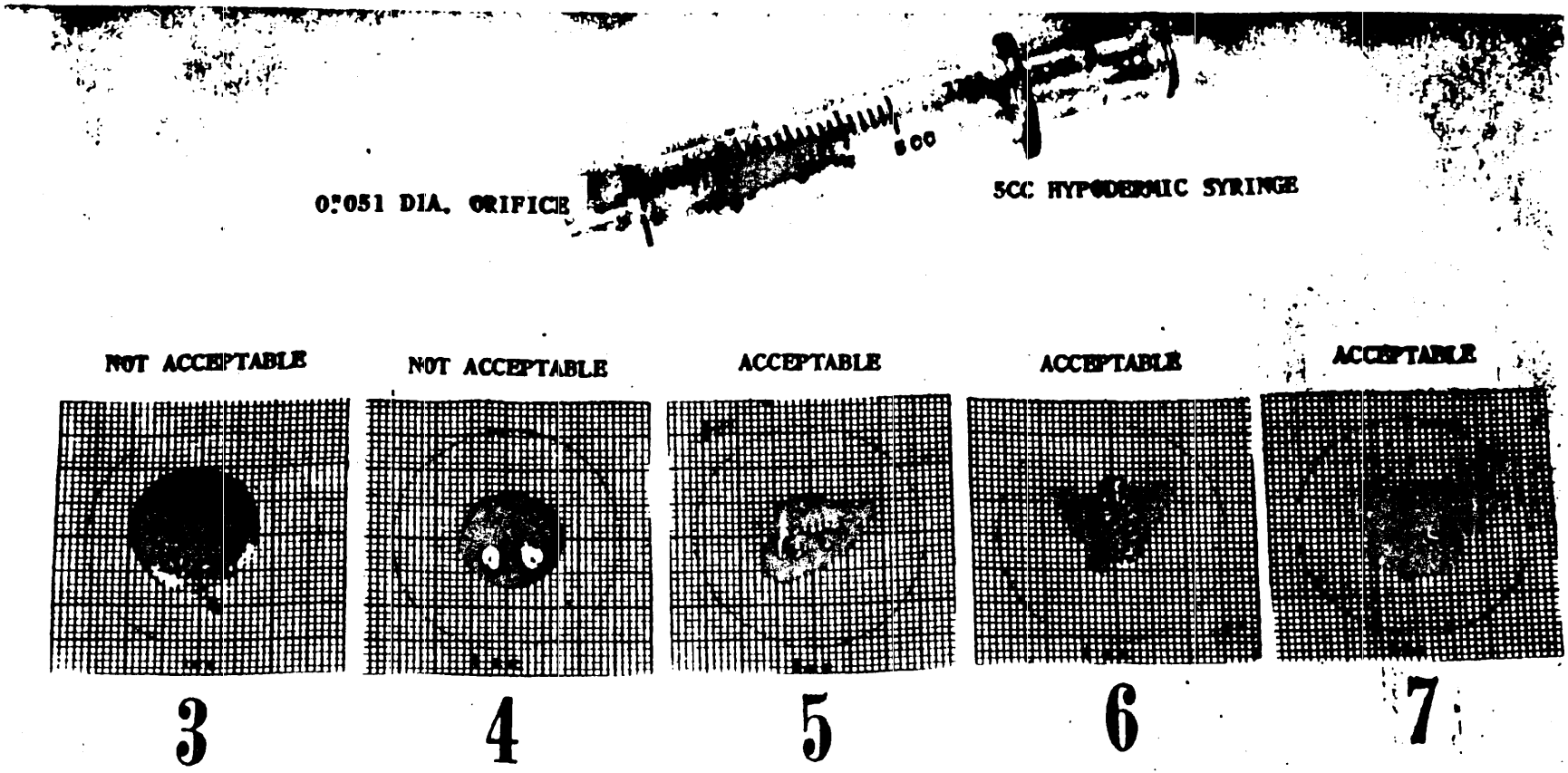


FIG. 4 ACCEPTANCE CRITERIA FOR VISCOSITY MEASUREMENTS

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4.4.3.2.2 Machining of Part B.- Part B shall be machined to dimensions indicated in Figure 2 from a glass-fiber base laminate epoxy resin panel meeting the requirements of MIL-G-21729A (Wep). Machining should be done either by grinding or with carbide tipped tools. All machining should be done dry and care taken to insure that the surface is not damaged by overheating or contaminated with lubricants. The initial finish on the plastic surfaces to be bonded shall be 80 microinches or better.

4.4.3.2.3 Surface preparation of Part A.- The minimum acceptable method for the preparation of metallic Part A, in lieu of chemical treatment, shall consist of sanding or sandblasting the faces to be bonded, followed by a vapor degrease in trichlorethylene just prior to coating with adhesive and assembly.

4.4.3.2.4 Surface preparation of Part B.- The laminated adherend, Part B, shall be sanded or sandblasted on the faces to be bonded, given a quick dip in a non-crazing solvent and dried immediately with cheese cloth just prior to coating with adhesive and assembly.

4.4.3.2.5 Re-use of Part A.- Part A may be re-used several times before resurfacing by grinding, provided all traces of the previously used resin are removed by solvent cleaning, followed by sanding with a fine abrasive paper. The bonding face of the solvent cleaned and sanded Part A shall be free of nicks and flat with no rounded edges. The sanding operation shall consist of placing a sheet of fine abrasive paper (No. 400 A Silicon Carbide) on a flat surface and by hand giving the bonding face of Part A a few circular rotations (clockwise and counter-clockwise) on the abrasive.

4.4.3.2.6 Re-use of Part B.- Part B may not be re-used.

4.4.3.3 Bonding.- Coat one end of each of the metal adherends and two faces of the laminate with adhesive taking care to see that all mating surfaces are thoroughly wetted by the adhesive prior to assembly. Place the adhesive coated laminate between the coated steel adherends and assemble in an alignment fixture, Figure 3, or any other acceptable V type alignment fixture. Test specimens shall be assembled under a pressure of 4 psi with the ground sides of the specimen aligned in the V slot of the alignment fixture.

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4.4.3.4 Cure of bonded specimens.- The bonded specimens shall be cured by either of the following procedures:

(a) 2 hours at 160° F immediately after assembly plus a post cure of 72 hours at 73° F and 50 percent relative humidity or
(b) 7 days at 73° F, 50 percent relative humidity.

4.4.3.5 Test procedure.- The bonded specimens shall be tested in tension at a rate of grip separation of .05 inches per minute, maximum. Record the maximum load carried by the specimen at failure. Specimens that break at a flaw remote from the glue line shall be discarded and retest made.

4.4.3.6 Calculations.- Tensile strength shall be calculated by dividing the breaking load by the area of the bonded surface. This result shall be expressed in pounds per square inch and, if possible, reported to three significant figures. For each series of tests the arithmetic mean of all values obtained shall be calculated and reported as the "average value". The average value shall be a result of no fewer than five tests.

4.4.4 Viscosity.- The viscosity shall be determined as follows using a minimum of two specimens:

4.4.4.1 Apparatus.- Hypodermic syringe, Luer type, resistance glass, 5 cc capacity, graduated in 1/5 cc intervals, tip in center.

4.4.4.2 Resin mix.- Viscosity measurements shall be obtained from a 100 gram resin sample formulated as described in 4.4.2.

4.4.4.3 Procedure.- Fill hypodermic syringe with thoroughly mixed resin conforming to 4.4.4.2. Insert lubricated syringe plunger to 5 cc mark. Support hypodermic syringe in a vertical position such that the tip is 0.75 inch above and perpendicular to a flat surface. Extrude 1 cc of the resin mix through the 0.051 inch diameter orifice of the hypodermic syringe onto a flat surface. The specimen must be extruded within 15 minutes after the addition of catalyst to the resin mix. Each specimen must meet the viscosity requirement of 3.4.2 for acceptance. Material which flows to a smooth or dome shaped surface, as illustrated in Figure 4, shall be evidence of failure to meet viscosity requirements.

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4.5 Failure of the thixotropic adhesive to meet any of the requirements and tests of this specification shall be considered cause for rejection of the lot represented by the sample tested.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Levels A and B.

5.1.1.1 Unit package.- Ingredients of subject material shall be unit packaged in accordance with Military Specification MIL-P-3863.

5.1.2 Level C.

5.1.2.1 Ingredients of subject material shall be packaged in a manner to insure protection from contamination and leakage.

5.2 Packing.-

5.2.1 Exterior containers.- Ingredients of subject material, unit packaged when applicable, shall be packed in accordance with the requirements of Military Specification MIL-P-3863.

5.2.2 Level C.

5.2.2.1 Exterior containers.- Ingredients of subject material, packaged as specified in 5.1.2, shall be packed to afford protection against damage during direct shipments from the supply source to the first receiving activity for immediate use. Shipping containers shall comply with applicable carrier rules and regulations.

5.3 Marking.- In addition to the markings required by the contract or order, unit packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use.- The thixotropic adhesive is intended for use in bonding glass cloth-epoxy laminated rings to metal components of warhead assemblies and must bridge voids up to .020 inches when applied to vertical and horizontal surfaces.

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6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Applicable level of packaging and packing
c. Type and size of container

6.3 Suggested source of supply.- The following manufacturers have supplied ingredients used to compound an adhesive complying to this specification:

Ingredient	Trade Name	Source
Resin, Class 1 Resin, Class 2	Paraplex P43 Paraplex P13	Resinous Products Div. Rohm & Haas Co. 224 Washington Square Philadelphia, Pa.
Accelerator	Nuodex Cobalt	Nuodex Products Co. 342 Madison Ave. New York, N.Y.
Catalyst	Methyl-Ethyl Ketone Peroxide (Lupersol DDM)	Lucidol Division Wallace & Tiernan, Inc. 1740 Military Road Buffalo 5, New York
Thixotropic agent	Cab-O-Sil	Godfrey L.Cabot, Inc. 77 Franklin Street Boston 10, Mass.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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